Finn Mulligan

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Automating Empathy

Over the last 50 years, life expectancy in the United States has risen from approximately sixty eight years to nearly eighty (Geffen). This can be attributed in no small part to the integration of modern technologies into our national healthcare systems and practices. A better understanding of the concepts of DNA and RNA which underscore nearly all health applications, has allowed for a better fundamental picture of how to care for health. Developments in surgical practices and procedures have greatly improved their applicability and efficacy. Drug discovery and production at scale has provided an incredible shift in their prevalence and effectiveness. Vaccine research has made us more resistant and resilient against endemic disease. These technological advances dramatically changed the shape of medicinal practices, and in doing so have made great leaps in extending lives, and improving the quality of those lives.

Artificial Intelligence is poised to be the next big driver of medical change in the world. As a general purpose technology, like the steam engine, AI can be applied to an incredibly wide array of tasks, and in its general applicability promises society-wide influence. The healthcare system in America is far from perfect, and there are many areas which could benefit from increased efficiency or effectiveness. To begin this essay, we will take a look at some of the potential fields of application within the healthcare system, and the weight of those potential benefits.

Artificial Intelligence has the potential to push the boundaries of modern medicine. There are some things which human medical professionals are unable to do, which an AI system might be able to accomplish. For example, Google has developed an AI system which can detect kidney failure up to two days before it happens, which is a great boon to medicine considering that such conditions are usually observed after the injury has taken place (Tomašev). In this instance, the tool by Google is able to gather important information and process it in a way that human doctors are unable to, and in doing so create very meaningful benefits to the patient that would not be possible without this new technology. In fields like radiology, where technicians analyze scans from machines like X-rays, AI is similarly poised to provide insights unattainable by human healthcare workers. Machine learning is particularly well suited to the task of analyzing images that look largely similar to many ones within the training corpus, but with some minor variations. Radiology is largely defined by such tasks, where each X-ray of a chest for example, looks generally similar to another, but the minor differences between each contain an incredibly rich amount of information. So, the integration of AI tools into these processes is leading to better diagnoses, faster, for less cost.

Similarly, Artificial Intelligence is assisting with impactful developments in medical research. The now notorious Alpha-Fold is an AI tool that predicts the way that protein chains will fold upon each other in three dimensional space. Given some input chain of amino acids, Alpha-Fold returns a 3D model of its anticipated structure. Previously, it would take about one PHD study in order to map out the shape of a single protein. With Alpha-Fold over three hundred and sixty thousand unique structures have been mapped, and the original paper has been cited hundreds and hundreds of times (Kovalevskiy). This application of AI into healthcare saved a nearly incomprehensible amount of human effort and time, and in doing so has acted as a springboard to an incredible variety of medical research avenues. Other developments may well come from AI that accelerate our understanding of human health, and how to best ensure its longevity and quality.

Artificial Intelligence also has the potential to democratize medical knowledge and quality. There are many medical ailments to which we do have methods of abatement and healing, but where access to this care is limited by geographic or economic considerations. Expert medical professionals are finite, and so too is the support that they can provide. This is made even more true the more esoteric the malady a patient may be suffering. Artificial Intelligence is well positioned to increase the accessibility to these limited sources of knowledge and healing. Diagnoses that previously would have necessitated a consultation with an domain expert who may well charge inaccessible fees, might well be given reliably into the hands of patients through more generalized practitioners with access to AI tools. This upskilling of healthcare worker capacity could enable more robustness and capabilities within our healthcare systems.

Artificial Intelligence also holds the capacity to automate a good deal of rote paperwork tasks in hospital work - freeing up healthcare professionals to spend more time interfacing with patients. A large portion of the daily tasks of healthcare workers within hospitals and across the industry entails entering data and recording information into digital ledgers - creating and maintaining medical records. These tasks, to some degree, may be sped up and made more efficient by a variety of AI powered tools. Speech-to-text programs might replace the need for active human transcription, and large language models, LLMs, can synthesize down relevant information and automatically update digital files. In assisting healthcare workers with these tasks, AI can enable them to spend more time working directly with patients, or doing less rote and monotonous medical work. This effort can free up time and in doing so increase quality and scale of care.

Finally, Artificial Intelligence may be used in order to optimize the management of medical resources and patient care. Artificial Intelligence may be used in order to determine the short term staffing requirements of a hospital, or which patient might benefit the most from finite medical resources. AI can assist in a whole host of decision making processes in order to help the hospital save costs and operate as efficiently and effectively as it possibly can. Considering the chronic funding strain on the American healthcare system, an ability to save on operating costs might directly commute to an increased capacity for care.

These potential benefits are weighty, and therefore necessitate careful consideration. The state of the modern medical system in America is a far thing from being as good as it might be, and these tools may make significant progress towards bettering our capacity to care for the health of our people. Based on this information alone, it would seem deeply reasonable to attempt to accelerate the rate of the integration of this technology as quickly as possible. Considering the domain is healthcare, the moral impact may be saving the lives of thousands of human lives. Indeed, this seems to be the goal of the current Secretary of Health and Human Services, RFK Jr, who in his confirmation hearing said: “President trump is determined to end the hemorrhage of rural hospitals, and he has asked me to do that through use of AI, telemedicine, which, these are innovations that I saw the other day. Cleveland Clinic has developed an AI nurse that you cannot distinguish from a human being, that has diagnostics as good as any doctor" (Finnegan).

I will argue however, that this claim overstates the applicability and capacities of Artificial Intelligence in a way that is misleading and dangerous. While the potential benefits are indeed immense, the costs requires a thorough investigation, and as will be discussed in the rest of this paper, are similarly sizable. AI has the capacity to revolutionize the shape of the healthcare system, but does not and will not contain the required features to automate the process. Human beings are essential to healthcare work, and are irreplaceable in many key regards. Contemporary AI systems are unfit to replace human healthcare workers, and doing so will harm patients, degrade care, and erode public trust in medicine.

As we can see by looking at contemporary cases across America, modern healthcare AI systems are not generally ready for autonomous use in healthcare. Across the nation healthcare systems are already attempting to see what value they can get out of these tools. However, this integration is largely premature and dangerous. We will examine three facets of modern Artificial Intelligence which render it unsuitable for such work: the fundamental unreliability of hallucinations, the lack of transparency in operation, and their capacity to perpetuate harmful biases.

Hallucinations are a phenomena ubiquitous in modern AI where the models make up or 'hallucinate' information that is not relevant or correct in some given context. The term acts as a catch all for basically any behavior that is not what is being looked for. Given how modern AI is built upon the mechanized statistical analysis of incredibly large sets of data, errors come up.

There is no grounding connection to the truth in the world for these models, they are just incredibly good and making educated guesses. These guesses are remarkable for the average of their quality and relevance, but fundamentally they are nothing more than guesses, and they can be nothing more.

It is true that modern healthcare is rife with human error, but this does not excuse the integration of machines which bring these errors to an industrialized scale. For example, Whisper is a speech to text AI model developed by OpenAI, a company at the forefront of the development of Artificial Intelligence. This tool has already "been used to transcribe an estimated 7 million medical visits"(Burke). As discussed earlier in this paper, this automation of labor might well have freed up a large amount of time and energy from healthcare workers which might then have gone on to provide more productive care. However, this tool is also well known to hallucinate large chunks of text and insert them directly into the transcripts it is creating. This is a massive problem considering its extant documented use at scale. Text generated by Whisper is sitting within the medical documents of millions of people, and its veracity is unclear. The confusion and potential harm caused by the errors in these documents is still yet to be fully revealed, but it is safe to say that it is significant. These harms could have been fully avoided without the effort to integrate AI systems into applications to which they are not yet ready.

Artificial Intelligence systems operate as 'black boxes' which is antithetical to ethical healthcare practices. For some given input, AI systems will provide some output, but the path and reasoning that goes from A to B cannot be fully revealed or understood. This is what is meant by 'black box,' there can be no effective and absolute certainty in determining why some AI system returns the answer it does to some prompt. However, transparency and traceability are core features of the ethical medical process (Montemayor). If an LLM is asked to provide some diagnosis based on some given information on patient health and symptoms, based on how they currently work, the true reasoning that they perform in order to arrive at their conclusion cannot be attained. Perhaps the writer of the prompt, or even the developer themselves, can ask the LLM to trace the steps it took to arrive at its diagnosis, but this just kicks the can of uncertainty down the road. There will still be no way to guarantee that the response the model provided in justification is actually an accurate representation of the internal steps it took to come to its conclusion. All we can know of the response is that it looks very much like what was written in the corpus of training data after the authors of those texts were asked to provide their various reasonings. The ethical medical process necessitates that practitioners be able to create consistent and reliable justification for their decisions, and this is simply something outside of the capacities of modern Artificial Intelligence.

Contemporary AI is also well known to perpetuate the biases which can be found within its training data. Considering that medical records in the United States contain clear reflections of our longstanding discriminatory and biased practices, Artificial Intelligence trained on this data is sure to contain this information, and likely to take action informed by it. A 2016 study found that "40% of first- and second-year medical students endorsed the belief that 'black people's skin is thicker than white people's'"(Sabin). It would seem like this belief would be an antiquated relic, but in fact it is still alive and kicking within the minds of medical students. This bias is present in much more than just surveys on the subject - the impact of these beliefs is widespread and incredibly pernicious. It has been found that Large Language Models are influenced by these biases: "they underestimate pain among Black individuals in the presence of false beliefs”(Deb B). Yet LLMs are being integrated at scale into medical systems. Additionally, a medical tool called Optum has been used in millions of cases in the United States in order to determine which patients would benefit the most from increased medical care. This tool however uses previous cost of care in order to inform its decisions. Due to the fact that the care for black patients has been deeply influenced by the biases of practitioners, less money has been spent on them. This results in assessments such that “at a given risk score, Black patients are considerably sicker than White patients” (Obermeyer). In order for Black patients to receive the equivalent care provided to white patients, they must be significantly more ill. To push for the integration of these systems is to push for the automation of our discriminatory practices.

While the issues that drive these shortcomings may be ameliorated or indeed resolved with future development, there will persist meaningful barriers between AI systems and the capacity to provide healthcare “that you cannot distinguish from a human being”(Finnegan). These issues are centralized around the incapacity for Artificial Intelligence to take accountability or produce empathy. Both features are critical for the delivery of ethical healthcare, and both are fundamentally inaccessible to AI systems.

The ability to empathize is a crucial capacity which healthcare professionals need to do in order to perform many aspects of their work ethically. However, as argued by Montemayor et al: “Empathy is an in principle limit for AI.” They define empathy as “*imagining feeling* where the feeling is about something specific, that is the focus of another person’s emotion.” Empathy is an experience that people are moved to, not one that they can directly call on command. It demands practice and intentionality to make oneself open to an empathetic experience, but ultimately is an internal state grounded in our biological systems. Montemayor et al argue that the processes associated with making oneself open to resonating with another emotionally “cannot be reduced to representations of rules and their application to cases.” This means that the current architecture of AI systems will be in principle incapable of providing genuine empathy. Empathizing with another human being entails more than just the processing of data and the following of rules.

Perhaps some lesser or partial form of empathy can be generated by AI, but it will be incomplete and counterproductive. Montemayor et al. describe how AI might potentially be able to muster ‘cognitive empathy,’ a sense derived from observation of behavior and pattern matching capacities. They point out though that “psychopathic patients are very good at cognitive empathy while lacking completely the remorse associated with emotional empathy” (Montemayor). In this light, it would seem like we are attempting to save on labor costs by replacing real empathizing humans with unfeeling, potentially psychopath-like machines. This feels like a pretty clearly bad idea.

There are real studied benefits to the health of patients when they feel like they are being genuinely empathized with. However, the best AI can provide is a simulated facsimile of this empathy that is in fact counterproductive. Any ‘empathy’ that an AI produces will be fundamentally misleading because it implies to the recipient that there is some being out in the world that is having a genuine experience of care and concern in regard to them. This is misleading, and is likely to result in directly harmful results as people are forced to interface with this underwhelming and insufficient replacement for human care. “Machines don’t give a damn” (Haugeland), but giving a damn is one of the key tasks within the job of administering ethical healthcare. Therefore, the effort to replace human healthcare workers with AI will be a degradation of care and result in harm to the patients. This degraded care is likely to reduce the confidence of the public in the capacities and capabilities of their healthcare system, and further the cycle of decreased effectiveness.

Furthermore, Artificial Intelligence is unable to take responsibility, which is another key aspect of ethical healthcare. When a mistake is made it is the responsibility of the healthcare provider to be accountable to it and bear the burden of the consequences. Artificial Intelligence however, is fundamentally incapable of being responsible. Should harm come to a patient based on the influence of an AI system, where does the blame fall? The developers of the AI system? The medical practitioners who employed the system? The patient for consenting to the use of such a system? We cannot hold the systems themselves accountable for the actions they take, because we don't think they have the capacity to make their own moral judgements. This responsibility gap cannot be easily resolved, but it must be for the ethical use of AI systems in healthcare. While extant, harm may be enacted onto some patient, and there might be no party to effectively hold responsible. In a scenario where these systems continue to be deployed with these issues as they stand, we would create a world where we have some automated harm being enacted on our society, and no party to bear the burden and responsibility of mitigation.

Artificial Intelligence systems are not moral agents, and so will continue to lack the capacity to take responsibility for the foreseeable future. When we hold a person accountable for some action, we are judging them based on information we determine around their intentions, autonomy, and knowledge. Artificial Intelligence systems are not grounded in these same processes like us humans. They cannot have intentions because all they can do is respond to external prompting. They cannot truly hold knowledge, because they are incapable of actually understanding anything they do. They operate as algorithms, very complex ones, but fundamentally just machines that take some input to some other output. Like ants that have left a pattern of crumbs spelling in English “the ants are marching,” there is no actual understanding of what has been written. They have no autonomy and so they cannot truly choose to do anything. Their actions are programmatic responses and they contain no internal experience.

Based on this, they cannot become members of our moral community. An AI system will feel the same way about a patient that makes a full recovery as they do a patient who dies under their care. That is to say, they won’t feel any type of way about either, because they cannot feel anything. This is a fundamental barrier between AI and the capacity to ethically replace human healthcare workers. Empathizing with patients is an essential function of the jobs of healthcare workers, and it is fundamentally inaccessible to Artificial Intelligence systems, rendering them unfit for such work.

Despite their various key shortcomings these systems are still being pushed actively into healthcare situations where they are not suitable. This drive, championed by RFK Jr, is not coming from a place of concern for human health. This can be seen clearly by the disregard for their obvious lack of capacities in key functions. Indeed, we can see that many cases across the United states where these Artificial Intelligence tools are being pushed into the healthcare system in order to save on costs. In his confirmation hearing, that is how RFK Jr. advertised the capacities of these systems: “as good as any doctor”(Finnegan). But incredibly crucially, they are not. Pushing this perspective cannot be sourced for a genuine concern for the health of our society.

Nurses across the nation have been forced to articulate the value of their labor, and push back against the integration of this technology. Michelle Mahon of National Nurses United has been quoted as saying: “the entire ecosystem is designed to automate, de-skill and ultimately replace caregivers”(Perrone). The administrative forces cramming these tools into hospitals display not only their disregard for the quality of care given to their patients, but also their disregard for their own workers. Human workers are expensive, and businesses are incentivized to optimize their profits, but the influence of this ideology in this application is particularly upsetting. It is the literal wellbeing of the people of our nation that we are letting become a secondary priority in the decision making processes around the future of healthcare. Many of the people in charge of this integration point to the potential benefits in a way that is fundamentally dishonest. Given the clear and well articulated shortcomings of these systems the frenzy behind the contemporary push towards integration can only be explained by greed and a lack of care for human wellbeing.

These qualities seem confirmed by the other efforts of RFK Jr. in reshaping the American healthcare system. In his service as Secretary of Health and Human Services, he has moved to shrink the size and budget of the already strained healthcare system. “‘We’re going to do more with less,’ Mr. Kennedy said, even as he acknowledged that it would be ‘a painful period for H.H.S’”(Jewett). Right there is both the acknowledgement of the harm his actions will cause and an obvious declaration of his lack of care in actually performing the responsibilities of his position. This idea that more can be done with less seems driven in no small part by his overconfidence in the applicability and capabilities of Artificial Intelligence systems. There is a clear lack of actual concern with the outcomes of these actions, and AI is being used as a justification for what is really an insidious and callused disregard for the wellbeing of the American people. Under his guidance over ten thousand federal healthcare workers have been fired (Jewett). This will harm the health of the population of America, and will increase the forces which pressure the healthcare systems towards premature Artificial Intelligence integration.

Artificial Intelligence has the capacity to bring incredible benefits to the healthcare system, but as tools in the hands of trained professionals, not as replacements for their labor. There truly are immense benefits which may be achieved through the use of this new technology. Development at the frontiers of medicine can be accelerated as seen by the benefits of Alpha-fold and diagnostic tools in the hands of trained professionals. Esoteric but valuable medical knowledge can be made more widespread and accessible to the masses. Rote tasks which consume large portions of medical professional's days might be relieved from them, allowing them to spend more time with patients. The business side of hospitals may be able to be optimized in a way that allows for increased quality and scale of care. In all these cases though, Artificial Intelligence serves as a method to enhance the quality and efficiency of care provided by human healthcare professionals. As stated by Cathy Kennedy, the president of the California Nurse Association: "Human expertise and clinical judgment are the only ways to ensure safe, effective, and equitable nursing care"(Perrone). This is a scenario in which it is integral that a human stays in the loop, and thus any effort to save on labor costs with the help of Artificial Intelligence is fundamentally misguided. Understandably misguided though, considering that there is explicitly misleading guidance coming from the position which is supposed to be trustable and authoritative on such matters: the Secretary of Health and Human Services.

To pursue the ethical integration of this technology there are several actions we can take guided by some logical priorities. First, we should seek to centralize transparency within the development and deployment of these systems. The developers of these systems should be accountable for clearly communicating the actual capacities and relevant shortcomings of their systems. There is a lot of mysticism and hand waving that AI developers like to lean into when advertising their products. It is easy for people to have unclear conceptions of these tools, as they are indeed capable, to some degree, of many tasks that previously were only accomplishable by humans. This inclines people towards an anthropomorphization that can lead them far away from an accurate and grounded understanding of what these models are really doing. Developers should be responsible for better communication around their products, as the lack of clear understanding in our society is enabling greed driven actors to cause harm.

Tied to this, we need to make sure that patients are aware of the extent to which Artificial Intelligence is being used in their treatment. There should be processes to ensure informed consent from the patients, as well as easy opt-out capacities. Patient autonomy needs to be respected and heeded, especially in contexts where there are noted risks and shortcomings of these systems as they are being applied. If a patient does not wish their data to be fed into these models, that should be achievable. If a patient does not wish for their treatment to be directed or influenced by these black box systems, then the hospital should be able to cater to that request.

Similarly, we need effective legislation to make sure that the development and deployment of these systems is done in a way that maximises their benefits while minimizing their harms. Currently, the burden of actually figuring out if these systems work or not is being shunted from the developers to the hospitals, and from the hospitals right onto the patients themselves. As articulated by Nurse Kennedy: “No patient should be a guinea pig and no nurse should be replaced by a robot.” It is the place of legislators to enforce this distinction, but there is currently an alarming paucity in this regard. It is indeed hard for legislators to keep up with the incredibly fast pace of technological development, but using that as an excuse for no effort is unacceptable. Legislation does work, and it is badly needed here, lest the capitalitic incentives outcompete the care for human wellbeing.

Relatedly, the public needs access to a better understanding of the actual capacities and limitations of these systems. It is a hard task to gain a full and accurate understanding of the contemporary technological scene, made much harder by the rapid pace of development. That said, the task should not be made harder by the seemingly semi-intentional obfuscation of their capacities by the stakeholders who stand to capitalize on the integration of this tech. There is nothing magic or incomprehensible about how Artificial Intelligence works, but it will take concerted effort to bring a more accurate and honest depiction of what they are into the minds of the general public.

Finally, we should push to centralize the voices of the patients and healthcare workers who are being affected in this discussion. Right now, the parties with the most say in these decisions are not the ones who are bearing the impact and repercussions of the integration. In the development of policy to guide this development, we should make sure to listen and respond to the actual people who are being impacted.

Artificial Intelligence has incredible capacity to change the way we take care of our people. We need to make sure that this impact is driven by genuine human concern for our conspecifics, rather than corporate greed or a political agenda. This effect can be achieved if we are able to be empathetic ourselves and make decisions driven by this concern and care. This however is not the guaranteed path and as such it will take concerted effort and attention in order to achieve. The idea that there exists out in the world some AI nurse “indistinguishable” from human caregivers is both misinformed and dangerous. Efforts towards integration guided by this fallacious belief will ultimately: cause harm to patients, filling their medical records with junk and providing unverifiable diagnosis, degrade the care provided by healthcare professionals, as they are instructed to differ their expertise and empathy to machines, and will reduce public trust in healthcare institutions, as they become less effective due to the burdens of this premature technological integration. Artificial Intelligence has immense potential for good, but only when understood properly, as it is unethical to attempt to automate empathy.

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